BSCOOK SECURITY LABS OFFICIAL PROTOCOL AUDIT

Report ID: BSC-SEC-2026-V3
Date: October 2026
Classification: Public Release

Smart Contract Security Audit & Verification Report

Comprehensive In-Depth Security Assessment, Formal Verification & Automated Static Analysis for the **BSCook Protocol** deployed on BNB Smart Chain (BSC).

98 / 100	0	0	PASSED
SECURITY SCORE	CRITICAL FINDINGS	HIGH FINDINGS	AVENGERDAO READY

1. Executive Summary

This official security audit was conducted by the **BSCook Protocol Security & Research Team**. The evaluation focused on formal verification, identification of potential attack vectors, economic vulnerabilities, reentrancy vulnerabilities, logic errors, access control soundness, and compliance with BNB Smart Chain standards.

Final Assessment Result: APPROVED / LOW RISK (Score: 98/100). No critical or high-severity vulnerabilities were identified. The smart contracts strictly adhere to OpenZeppelin standards, utilize multi-signature threshold authorization, employ non-custodial liquidity locking mechanisms, and contain no rug-pull, honeypot, or arbitrary fund draining logic.

2. Project & Assessment Overview

Project Name	BSCook Protocol	**Deployment Network**	BNB Smart Chain (BSC Mainnet - Chain ID: 56)
Official Domain	https://bscook.fun	**Architecture**	Bonding Curve Launchpad + Uniswap V4 LP Locker
Solidity Compiler	v0.8.20 (viaIR: true, runs: 800)	**Primary Framework**	Hardhat / OpenZeppelin v5.x

3. Scope of Assessed Contracts

Contract Name	Mainnet Contract Address	Status
BSCookFactory	`0xe8a0beEEdd39af3a87De11d6b120D0fFa874867D`	Verified Exact Match
BSCookToken (Impl)	`0x66aEfEE3B1f3163f036Aeee733035df09F49Fc02`	Verified Exact Match
BSCookLocker	`0x3a8587057f85eb318F52c2C9ae148940741CFCfD`	Verified Exact Match

Contract Name	Mainnet Contract Address	Status
BSCookTimelock	`0x4cec2C1DA4D2f460402aC38aF04Ed29D0C922c83`	**Verified Exact Match**
BSCookMultiSig	`0xAa2699E1737052869c97a681d3d6d1C9b969120e`	**Verified Exact Match**
ContractValidation	Internal Shared Safety Library	**Verified Exact Match**

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4. Core Security Checks & Vulnerability Analysis

Security Invariant	Verification & Mitigations Tested	Result
Reentrancy Attack	All external mutating routines protected with OpenZeppelin `ReentrancyGuard`. CEI (Checks-Effects-Interactions) strictly observed.	SECURE / PASSED
Honeypot & Blacklist Risk	Source code contains NO blacklist mechanisms, NO trading pause for users, and NO hidden dynamic transfer taxes. Tokens trade friction-free.	SECURE / PASSED
Arbitrary Token Minting	Tokens deployed via `BSCookToken` have deterministic initial supply created only on initialization. No unrestricted mint function exists.	SECURE / PASSED
Rug-pull & LP Theft	LP positions resulting from bonding curve graduation are minted and directly deposited into `BSCookLocker` with immutable locking locks.	SECURE / PASSED
Centralization & Governance	Administrative actions require consensus via `BSCookMultiSig` combined with execution delays enforced by `BSCookTimelock`.	SECURE / PASSED
Arithmetic & Overflow	Built with native Solidity 0.8.20 checked arithmetic; bonding curve calculations validated against boundary and precision-loss limits.	SECURE / PASSED
DEX Integration	Safe interaction with Uniswap V4 PoolManager singleton, hook verification, and Permit2 standard compliance.	SECURE / PASSED

5. Findings Summary & Remediation Tracking

ID	Severity	Description	Resolution Status
`BSC-01`	LOW	Enforce explicit minimum bounds on Timelock delay parameters.	RESOLVED (In-code checks applied)
`BSC-02`	INFORMATIONAL	Gas optimization via compiler `viaIR` and 800 optimizer runs verified.	CONFIRMED OPTIMAL

6. Methodology & Verification Tools

The audit followed a multi-stage process incorporating static analysis tools (**Slither 0.10.x**, **Mythril**, **Solhint**), manual architectural review, invariant fuzzing, and bytecode verification against BNB Smart Chain Mainnet deployment artifacts. All identified bytecode matches the published source code bit-for-bit.

7. Protocol Verification Commitment

The BSCook Core Security & Engineering Team commits to ongoing security monitoring, transparent smart contract verification on BscScan, and adherence to AvengerDAO security standards. All protocol contracts remain immutable or protected by multi-signature and timelock delays.